A FINITELY PRESENTED MODULE FLAT ALONG A FIBRE BECOMES PROJECTIVE OVER AN AFFINE ETALE NEIGHBOURHOOD

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The pointwise projective version of the flat along the fibre statement. Same chapter flat.tex, label proposition-finite-presentation-flat-at-point, invoked at proof line 3368. Produces, for each point of the fibre, elementary etale neighbourhoods of the base and of the source whose global sections of the pulled back module are projective over the ring of the base neighbourhood; assembling finitely many of them is what the counted proof supplies.
- Existence of an elementary etale neighbourhood dominating finitely many given ones. More on Morphisms, label more-morphisms-lemma-elementary-etale-neighbourhoods, invoked at proof line 3387. Gives the single dominating elementary etale neighbourhood of the base, which may also be assumed affine, that replaces the finitely many pointwise neighbourhoods.
- Projectivity ascends along a base change of rings. Algebra, label algebra-lemma-ascend-properties-modules, invoked at proof line 3400. Shows that the direct sum of the base changes of the pointwise projective modules is again projective over the ring of the new base.

Source reference. Original lemma 05A3: sources/stacks-chow/tags.txt, source lines 6836–6836. Navigation pointer for \ref{algebra-lemma-ascend-properties-modules} used inside the counted span (line 3400); the argument that consumes it is retained in the span.

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Source reference. Original lemma 057B: sources/stacks-chow/tags.txt, source lines 6739–6739. Navigation pointer for \ref{more-morphisms-lemma-elementary-etale-neighbourhoods} used inside the counted span (line 3387); the argument that consumes it is retained in the span.

Source reference. Original proposition 05ID: sources/stacks-flat/pinned-original-source.tex, source lines 3287–3287. Navigation pointer for \ref{proposition-finite-presentation-flat-at-point} used inside the counted span (line 3368); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent sheaf on X. Let $s \in S$. Assume that*

(1) *f is of finite presentation,*
(2) *$\mathcal{F}$ is of finite presentation, and*
(3) *$\mathcal{F}$ is flat over S at every point of the fibre X_s.*

Then there exists an elementary étale neighbourhood $(S', s') \to (S, s)$ and a commutative diagram of schemes

$$\begin{array}{ccc} X & \xleftarrow{g} & X' \\ \downarrow & & \downarrow \\ S & \longleftarrow & S' \end{array}$$

such that g is étale, $X_s \subset g(X')$, the schemes X', S' are affine, and such that $\Gamma(X', g^*\mathcal{F})$ is a projective $\Gamma(S', \mathcal{O}_{S'})$-module.

Proof. For every point $x \in X_s$ we can use Proposition 05ID to find a commutative diagram

$$\begin{array}{ccc} (X, x) & \xleftarrow{g_x} & (Y_x, y_x) \\ \downarrow & & \downarrow \\ (S, s) & \longleftarrow & (S_x, s_x) \end{array}$$

whose horizontal arrows are elementary étale neighbourhoods such that Y_x, S_x are affine and such that $\Gamma(Y_x, g_x^*\mathcal{F})$ is a projective $\Gamma(S_x, \mathcal{O}_{S_x})$-module. In particular $g_x(Y_x) \cap X_s$ is an open neighbourhood of x in X_s. Because X_s is quasi-compact we can find a finite number of points $x_1, \ldots, x_n \in X_s$ such that X_s is the union of the $g_{x_i}(Y_{x_i}) \cap X_s$. Choose an elementary étale neighbourhood $(S', s') \to (S, s)$ which dominates each of the neighbourhoods (S_{x_i}, s_{x_i}), see More on Morphisms, Lemma 057B. We may also assume that S' is affine. Set $X' = \coprod Y_{x_i} \times_{S_{x_i}} S'$ and endow it with the obvious morphism $g : X' \to X$. By construction $g(X')$ contains X_s and

$$\Gamma(X', g^*\mathcal{F}) = \bigoplus \Gamma(Y_{x_i}, g_{x_i}^*\mathcal{F}) \otimes_{\Gamma(S_{x_i}, \mathcal{O}_{S_{x_i}})} \Gamma(S', \mathcal{O}_{S'}).$$

This is a projective $\Gamma(S', \mathcal{O}_{S'})$-module, see Algebra, Lemma 05A3. □